

2022-Ph-H1-Q20

Section: Particles and Waves

Topic: Wave-Particle Duality

What is the evidence for the wave nature of electrons?

Answer: B

(Diffraction pattern)

Guidance for Students:

Particles like electrons can show wave behaviour — they **diffract** and interfere, forming patterns like light waves.

Revision Tips:

- Electron diffraction = wave-like behaviour.
 - Supports de Broglie hypothesis: $\lambda = \frac{h}{p}$
 - Only observed when electrons pass through narrow gaps/slits.
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